

Statement of Purpose - Research Internship at ILPRL Lab

I am highly motivated and have strong interest to join Nepal's top notch AI research lab in the field of Natural Language Processing to build technologies and research on development projects that can possibly help in digitizing resources for underrepresented languages and communities with low resources in NLP.

As a Data Science Student, I have developed a strong foundation in Python, Data Analysis and Machine Learning Concepts. I have built several Machine Learning Models and created proper modular NLP pipeline and other NLP related projects some in ideation and some in development phases, which have strengthened my problem-solving skills and technical skills.

I have had keen interest in NLP related tasks since the start of my tech journey, I have been closely following the work of ILPRL lab and am consistently inspired by the impactful research and development initiatives led by Respected Bal Krishna Bal Sir. I was particularly impressed by ILPRL's work on advancing Nepali NLP with transformer Based Pre trained models, where approx. 28 GB of Nepali corpus was developed, and beating existing models and surpassing previous benchmarks of models like distilgpt-nepali, setting a new standard for low resource language research like Nepali addressing its significant research gap.

In addition, I am currently working on Nepali Spellchecker project, where I have made progress in improving the Hunspell-based system, I am eager to further contribute and collaborate with ILPRL lab to enhance and develop tools for Nepali Language Community. I believe ILPRL lab perfectly provided the ideal platform for such projects and other projects like Multilingual Machine Translation, RAG based information Retrieval, Nepali LLM development.

Thank you for considering my application.